Biochemistry Cume
November 15, 2002

1. (10 points) All transfer RNA molecules terminate with the base sequence CCA at their 3' ends.
   a. Predict whether the secondary structure of this sequence will be single-stranded or a portion of a double-stranded stem of the transfer RNA.
   b. Justify your answer to part a.

2. (20 points) Draw a chemical structure for the last 3 phosphoribose units at the 3' end of a molecule of transfer RNA. (Use just the one-letter abbreviations for the aromatic bases, but show all other atoms.)

3. (10 points) A transfer RNA molecule is found to be incomplete or damaged at its 3' end so that its 3'-terminal base sequence is only CC. What functional problem will this create for a living cell?

4. (5 points) An enzyme is found to repair the incomplete transfer RNA molecule problem 3. What substrate (in addition to the tRNA molecule) will be suitable for the repair process? Why?

5. (2 points) Suppose that the repaired intact transfer RNA molecule in question 4 is a cognate, isoaccepting transfer RNA for methionine.
   a. (1 point) Define cognate.
   b. (1 point) Define isoaccepting.

6. (33 points) Suppose that the repaired intact transfer RNA molecule that results in question 4 is a cognate, isoaccepting transfer RNA for methionine. An enzyme attaches methionine to the transfer RNA.
   a. (3 points) Name the enzyme.
   b. (5 points) What cofactor is needed for the reaction?
   c. (10 points) What 2 functions are provided by the cofactor?
   d. (15 points) Draw a properly bonded structure for the appropriate phosphoribose terminal of the product molecule, also including the structure for the attached methionine. (Use a one-letter abbreviation for the aromatic base.)

7. (20 points) The molecule that results from problem 6 is treated with aqueous NaOH. Draw any three products that result.
   (Do not include H_2O or Na^+ among your answers.)